

Highlights

- Debiased graph propagation is biased beyond one hop because walks repeat edges, and we derive this bias exactly for inverse-propensity and doubly robust estimates.
- DRUP removes the repeated-walk bias for any odd number of hops, at the cost of the uncorrected three-hop operator.
- On a log with known exposure, the correction removes an 18–29% overestimate of an allocation’s utility and cuts three-hop error 4 to 12 times.
- The guarantee fails when degree weights come from the log, a condition that standard cross-fitted debiasing pipelines violate; we locate the cause by simulation.
- Across Coat, Yahoo! R3, KuaiRand and KuaiRec and up to 26 tuned baselines, the correction leaves three-hop accuracy unchanged within 0.0014 nDCG.